

CO4_AT1_DATA INTERPRETATION**1. Real-Time Detection System**

A surveillance system uses the following techniques:

Method	Accuracy	Speed
Viola-Jones	78%	Fast
YOLO	92%	Moderate
Deep Learning	95%	Slow

Compare the three techniques and evaluate which method should be selected for a surveillance system operating under strict latency constraints while maintaining acceptable accuracy. Justify your decision considering deployment conditions.

The three techniques can be compared based on their accuracy and processing speed. Viola-Jones provides an accuracy of 78% and has fast processing speed. It is suitable for systems with limited computational resources and very strict latency requirements, but its lower accuracy may result in incorrect detections.

YOLO provides 92% accuracy with moderate processing speed. It offers a good balance between detection accuracy and real-time performance.

Deep Learning provides the highest accuracy of 95%, but its processing speed is slow. Therefore, it may not be suitable for a surveillance system where immediate detection is required.

For a surveillance system operating under strict latency constraints while maintaining acceptable accuracy, YOLO is the most suitable choice. It provides high detection accuracy and acceptable processing speed. However, if the deployment environment has very limited hardware resources and requires extremely fast processing, Viola-Jones can be considered.

Conclusion: Therefore, YOLO is recommended because it provides the best balance between high accuracy and real-time performance.

2. Robust Object Detection

Method	Normal	Occlusion	Low Light
Viola-Jones	80%	60%	55%
YOLO	90%	75%	70%
Deep Learning	95%	85%	80%

Compare the robustness of the three methods under normal, occluded, and low-light conditions. Evaluate which method provides the best overall performance and discuss the trade-offs involved.

The robustness of the three methods can be compared under normal, occlusion, and low-light conditions.

Under normal conditions, Viola-Jones achieves 80% accuracy, YOLO achieves 90%, and Deep Learning achieves 95%. Deep Learning provides the highest performance.

Under occlusion, the accuracy of all methods decreases. Viola-Jones achieves 60%, YOLO achieves 75%, and Deep Learning achieves 85%. Deep Learning is more capable of detecting partially hidden objects.

Under low-light conditions, Viola-Jones performs the lowest with 55% accuracy, while YOLO achieves 70% and Deep Learning achieves 80%.

Overall, Deep Learning provides the best performance under all conditions. It is the most robust method because it maintains high accuracy even when objects are occluded or lighting conditions are poor.

However, the trade-off is that Deep Learning requires more computational resources and may have slower processing speed. YOLO provides a good compromise between robustness and real-time performance, while Viola-Jones is faster but less accurate under difficult conditions.

Conclusion: Deep Learning is the best method for overall robustness, while YOLO is preferred when real-time processing is also important.

3. Optical Flow Analysis

Two optical flow methods are available:

- **Method A: Produces stable motion vectors but misses small motions.**
- **Method B: Detects fine motion but produces noisy vectors.**

Compare the two methods and evaluate which method is more suitable for motion tracking in dynamic scenes. Justify your answer based on accuracy, stability, noise sensitivity, and application requirements.

Method A and Method B have different advantages and limitations. Method A produces stable motion vectors, which makes the tracking results smooth and reliable. It is also less sensitive to noise. However, its main disadvantage is that it may miss small or fine movements.

Method B can detect fine and small motions, making it more accurate for detailed motion analysis. This is useful in dynamic scenes where objects move in different directions and at different speeds. However, Method B produces noisy motion vectors, which can reduce the stability and reliability of the tracking results.

For motion tracking in dynamic scenes, Method B is generally more suitable because detecting fine motion is important. Its noise problem can be reduced by applying filtering or smoothing techniques.

If the application requires stable tracking and is less concerned with small movements, Method A would be a better choice.

Conclusion: Method B is recommended for dynamic scenes because of its ability to detect detailed and fine motion, provided that appropriate noise reduction techniques are used.

4. Model Generalization and Overfitting – 10 Marks

Model	Training Accuracy	Testing Accuracy
A	98%	70%
B	90%	88%

Compare the two models and evaluate which model is more reliable for real-world deployment. Explain the generalization ability and overfitting risk of each model and justify your selection.

Model A has a training accuracy of 98% but a testing accuracy of only 70%. The difference between the training and testing accuracy is 28%. This large difference indicates that Model A is suffering from overfitting. The model performs very well on the training data but does not generalize well when tested on new or unseen data.

Model B has a training accuracy of 90% and a testing accuracy of 88%. The difference is only 2%, which indicates that the model performs consistently on both training and testing data.

Therefore, Model B has better generalization ability and a lower risk of overfitting. Although Model A has higher training accuracy, its poor testing accuracy makes it less reliable for real-world applications.

Conclusion: Model B is more reliable for real-world deployment because it generalizes well to unseen data and has a lower risk of overfitting.

5. System Performance under Constraints

Model	Accuracy	Latency	Cost
A	88%	50 ms	Low
B	92%	80 ms	Moderate
C	95%	120 ms	High

Compare the three models and evaluate which model provides the best balance between accuracy, latency, and cost for a real-time application. Justify your choice considering practical deployment constraints.

The three models are compared based on accuracy, latency, and deployment cost.

Model A has an accuracy of 88%, latency of 50 ms, and low cost. It is the fastest and least expensive model, but it has the lowest accuracy.

Model B provides 92% accuracy with 80 ms latency and moderate cost. It offers a good balance between performance, response time, and deployment cost.

Model C provides the highest accuracy of 95%, but it has the highest latency of 120 ms and high deployment cost. This may make it less suitable for applications that require fast real-time responses.

For a practical real-time application, Model B provides the best balance. It achieves high accuracy while maintaining acceptable latency and moderate deployment cost. Model A can be selected when low cost and very fast response are the main requirements, whereas Model C is suitable when maximum accuracy is more important than latency and cost.

Conclusion: Model B is recommended because it provides the best balance between accuracy, latency, and cost for real-time deployment.